

The empirical recurrence for OEIS A297400: recovering a conjecture that is not written down

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Abstract

OEIS A297400 records “Empirical recurrence of order 69”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 118$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 69, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 69 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A297400 is “Number of $7 \times n$ 0..1 arrays with every 1 horizontally, diagonally or antidiagonally adjacent to 1 neighboring 1.”. It has offset 1 and begins

1, 178, 963, 3924, 74736, 640562, 4279111, 46261441, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Dec 29 2017 11:01:31, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 16384$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 118$ of the 16384, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 118$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 236$ exact terms of the model returns order 69 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	1	c_{19}	−58713190409	c_{37}	−213993087329800	c_{55}	12688554968002
c_2	11	c_{20}	−174847970394	c_{38}	226379360382451	c_{56}	1115804324443
c_3	593	c_{21}	462739053610	c_{39}	337918792155558	c_{57}	−1866317345334
c_4	822	c_{22}	790169719402	c_{40}	−9982882599156	c_{58}	−1867117807450
c_5	−129	c_{23}	1223095569343	c_{41}	−342370275425640	c_{59}	−413431970440
c_6	−93445	c_{24}	−1534625245118	c_{42}	−204602400784707	c_{60}	331501182544
c_7	−164935	c_{25}	−4947042703597	c_{43}	187541437098165	c_{61}	198363149900
c_8	−120791	c_{26}	−5500345834685	c_{44}	310109577567327	c_{62}	45943725652
c_9	6618061	c_{27}	−2047636439757	c_{45}	26820646178857	c_{63}	−26082650728
c_{10}	8795039	c_{28}	795554251312	c_{46}	−150311730919388	c_{64}	−12586924776
c_{11}	11666129	c_{29}	41414626633899	c_{47}	−151686738658693	c_{65}	−2147785984
c_{12}	−220758220	c_{30}	33337936763376	c_{48}	19278808578488	c_{66}	563709984
c_{13}	−194363558	c_{31}	−75845465869095	c_{49}	107544633296485	c_{67}	290267712
c_{14}	−639277599	c_{32}	−114183587379628	c_{50}	28462503903431	c_{68}	47892480
c_{15}	4249790054	c_{33}	30223963838980	c_{51}	−4483157304681	c_{69}	9907200
c_{16}	3016176097	c_{34}	263870471800163	c_{52}	−50684737287517		
c_{17}	13526672175	c_{35}	120265827740622	c_{53}	−593549694830		
c_{18}	−54247401308	c_{36}	−292086807702297	c_{54}	3967835186111		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 69, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $69 + 4$ terms removed returns order 69 as well, so $\deg q_{\min} = 69 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $69 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 69 that this sequence satisfies — and that family turns out to have exactly one member.

References

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